

Supplemental Readings – MA206

These readings cover a small number of topics the course assesses but Devore's *Probability and Statistics for Engineering and the Sciences* (9th ed.) does not treat directly. Each is written to match the text's terminology and level, and each is tagged with the lesson it supports (for example, **S1**). Read the tagged supplement alongside the assigned Devore sections for that lesson.

S1 · Levels of Measurement

Supports Lesson 1 (Types of Data & Study Design; read with Devore, Section 1.1).

Devore classifies data as **categorical** or **numerical** (Section 1.1). A complementary classification, a variable's **level of measurement**, records how much information its values carry, and therefore which summaries are meaningful. There are four levels: the first two refine Devore's *categorical* type, the last two his *numerical* type.

- **Nominal:** unordered categories that serve only as labels (for example, branch).
- **Ordinal:** ordered categories whose spacing is unequal or undefined (for example, class year).
- **Interval:** numerical with equal spacing but no true zero, so ratios are not meaningful (for example, temperature in °F).
- **Ratio:** numerical with equal spacing and a true zero, so both differences and ratios are meaningful (for example, height or elapsed time).

The level fixes the appropriate summary: a **mode** applies at any level, a **median** requires at least ordinal data, and a **mean** and **standard deviation** require at least interval data.

S2 · The $1.5 \times \text{IQR}$ Rule for Identifying Outliers

Supports Lesson 2 (Measures of Location & Variability; the fourth spread is Devore, Section 1.4).

Devore measures spread in the middle of a data set with the **fourth spread** f_s , the difference between the upper and lower fourths (Section 1.4). In most software, and in this course, we use the equivalent and more familiar names: the **first and third quartiles**, Q_1 and Q_3 , and the **interquartile range**,

$$\text{IQR} = Q_3 - Q_1,$$

which is the same quantity as f_s . The IQR is the width of the middle 50% of the data and, unlike the range or the standard deviation, is resistant to extreme observations.

A common convention for flagging unusually large or small values uses the IQR. Compute the **fences**

$$\text{lower fence} = Q_1 - 1.5 \text{IQR}, \quad \text{upper fence} = Q_3 + 1.5 \text{IQR}.$$

Any observation below the lower fence or above the upper fence is **flagged as an outlier**. (A value more than 3 IQR beyond a quartile is sometimes called an *extreme* outlier, and one between 1.5 and 3 IQR a *mild* outlier.) These fences are also where a boxplot ends its whiskers.

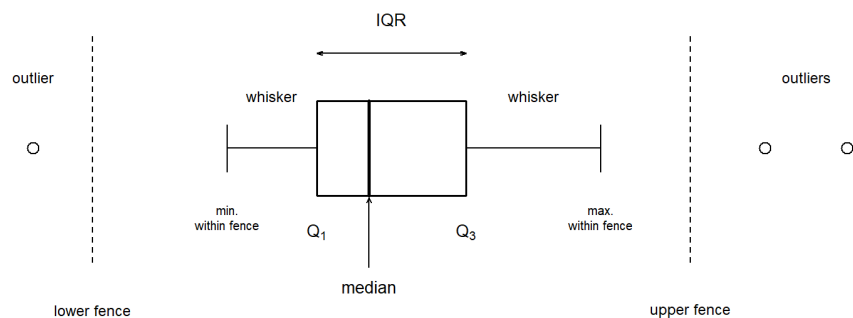


Figure 1: The $1.5 \times \text{IQR}$ fences: any point beyond a fence (open circles) is flagged as an outlier.

The rule is a **screening tool, not a verdict**: a flagged point is a candidate for a closer look, not automatically an error, and the multiplier 1.5 is Tukey's convention rather than a law of nature.

S3 · Correlation vs. Causation

Supports Lesson 14 (Exploratory Data Analysis; sample correlation is formally developed in Devore, Section 12.5).

A sample correlation coefficient r close to ± 1 , or a strong linear pattern in a scatterplot, shows that two variables tend to move together. It does **not**, by itself, show that one variable causes the other. This distinction is central to interpreting an exploratory analysis honestly, and it is the single most common error in reading a statistical study.

Two variables can be strongly associated for any of several reasons:

- **Direct causation:** changes in the explanatory variable genuinely produce changes in the response.
- **Reverse causation:** the presumed effect is in fact the cause.
- **A lurking variable:** a third variable influences both, producing an association between them even though neither causes the other.
- **Coincidence:** in a small sample, a large r can arise by chance alone.

A **lurking variable** (or **confounder**) is a variable, not among those under study, that is associated with both the explanatory and the response variable and can account for some or all of their observed association.

Establishing causation requires more than a correlation. The strongest evidence comes from a **randomized, controlled experiment** (Devore, p. 10), in which the explanatory variable is deliberately manipulated and other factors are held fixed or randomized across groups. In an **observational study** (Devore, p. 365; the usual situation for the course project) we can measure and describe association, but a causal claim must be made cautiously and must confront the plausible lurking variables.

When you report a correlation in an exploratory analysis, state the **strength** and **direction** of the linear association, then note explicitly that association alone does not establish causation and identify at least one plausible lurking variable.

S4 · Coefficient of Determination: r^2 and R^2

Supports Lesson 33 (Multiple Linear Regression II; R^2 is developed in Devore, Section 13.4, and the simple-regression r^2 in Section 12.5).

In simple linear regression Devore writes the **coefficient of determination** in lowercase, r^2 (Section 12.5); in multiple regression the same idea appears in uppercase, R^2 (Section 13.4). The two measure exactly the same thing, and the change in case is not arbitrary.

Both are defined the same way, from the regression sums of squares:

$$r^2 = R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = \frac{\text{SSR}}{\text{SST}},$$

the proportion of the total variation in the response y that the fitted model explains. A value near 1 means the model accounts for most of the variability; a value near 0 means it accounts for little.

The notation differs because of how many predictors are involved. In **simple** linear regression there is a single predictor x , and the coefficient of determination is exactly the **square of the sample correlation coefficient** r between x and y , hence lowercase r^2 . In **multiple** regression there are several predictors, so no single correlation r exists; the uppercase R^2 is the **coefficient of multiple determination**, and its square root R is the **multiple correlation coefficient**, the correlation between the observed values y and the model's fitted values $\hat{y}$.

One caution specific to R^2 : **adding any predictor to a multiple regression can only increase R^2 , never decrease it**, even a predictor that carries no real information. To compare models with different numbers of predictors, use the **adjusted R^2** , which penalizes the model for each added predictor and can decrease when a useless predictor is included.

	Simple regression	Multiple regression
Symbol	r^2 (lowercase)	R^2 (uppercase)
Name	coefficient of determination	coefficient of multiple determination
Ties to correlation	$r^2 = (\text{sample correlation } r)^2$	$R = \text{corr}(y, \hat{y})$
Devore section	12.5	13.4